Geophys. J. R. astr. Soc. (1980) **62**, 577–590

A palaeomagnetic study of Cambrian red beds from Cartaret, Normandy, France

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Received 1979 October 10; in original form 1979 July 19

Summary. Results are reported from a short stratigraphic sequence of Lower Cambrian red sediments outcropping in Dennemount Quarry, Cartaret and Pointe-de-Rozel, Normandy. After af cleaning the data show a smear from steep to shallow positive magnetic inclination. Changes in inclination appear to be related to stratigraphic position – steep inclinations are stratigraphically lower than shallow inclinations. Further thermal and chemical treatment reveal that this smear does not record polar wander, rather it is the result of the interaction of two distinct remanence phases. Fold tests, chemical cleaning and opaque petrology indicate that the sequence of remanence acquisition was first the shallow south-westerly magnetization, followed later by the steep magnetization.

Pole FB (13° S, 301° E, $\alpha_{95} = 8°$) appears to date the initial depositional event. It may either be a chemical, or a detrital magnetization, or some combination of the two. Pole FA (22° N, 354° E, $\alpha_{95} = 11°$) is related to the red pigmentation of the sediments. Although no strict age constraints can be applied to this remanence, a basal Ordovician age is preferred.

Introduction

Although, faunal provincialism in the Ordovician had been known for some time, it was J. T. Wilson in 1966 who first interpreted the juxtaposition of distinct faunal realms as indicating that previous cycles of continental drift had been operative. Further detailing of faunal provinces by Whittington & Hughes (1972) and later by Burrett (1973) showed that the present European region contains a number of faunal provinces which at different times during the Ordovician were either isolated, or interacting with their immediate neighbours. For example, southern Britain during the Arenig to Llanvirn shows close affinities with southern Europe, while during the Tremadoc and after the Llanvirn it is more closely aligned with northern Europe. From the faunal evidence alone it is uncertain if these changes represent the movement of central Britain to and fro across a mid-European ocean or whether they record changes in climate, or ocean current, patterns. Resolution of this dichotomy is possible through a comparison of contemporaneous palaeomagnetic results from the different faunal provinces as they evolved through the Ordovician.

Presently available palaeomagnetic data permit the description of a general distribution of the major landmasses for certain intervals of the Lower Palaeozoic. However, the data base is presently insufficient to detect detailed motions between smaller continental fragments. Since microplate rotations may have been quite significant within the central European block (Badham & Halls 1975; Lorenz 1976), these are exactly the type of pole differences that we are hoping to detect.

With respect to the European region it is the Ordovician pole path relative to southern Britain that is by far the best defined. The Cambrian pole path relative to Britain is as yet uncertain, the only published poles from Cambrian sediments being considered questionable. Their proximity to the representative Carboniferous pole could be considered as indicating that the Cambrian sediments are carrying Carboniferous magnetizations. Up till recently Lower Palaeozoic data for central Europe were even more scarce – the only data available being that published by Bucha (1961, 1965) from the Ukrainian massif. A number of studies of this area have now been completed (Jones *et al.* 1979; Hagstrum *et al.* 1978) and it is to be expected that our understanding of the motions of the continental fragments trapped between the jaws of Laurentia and Gondwana will soon be improved. This study provides additional data points for the clarification of these motions.

Red beds often have complex magnetic histories, and can in idealized circumstances record a number of remanence directions corresponding to a pole path segment (Roy & Park 1974; Roy & Robertson 1978). The differing remanence directions may record the gradual transition of the red bed from a newly deposited sediment, through diagenesis, and finally to later metamorphism. No strict timetable can be erected to describe the temporal sequence of these events. Rather each deposit has its own unique character. Further the relative magnitude of the various remanence components varies not only between different sites in the same unit but also between various samples at the same site. However, if the true directions of the individual remanence phases can be isolated, and their sequence of acquisition established then red beds can provide detailed apparent pole path segments.

The Cambrian sediments at Cartaret provide an excellent section of palaeomagnetic analysis. The red beds at Dennemount Quarry from which most of the samples were collected contain hyolithids and sponge spicules suggestive of a Cambrian age (Doré 1963). The immediately overlying limestones which have a more complete fauna containing archaeocyathids and trilobites define the age of these beds as Lower Cambrian. The presence of dessication cracks and phosphate nodules in these beds suggests that they were probably deposited at fairly low latitudes. The beds at the second locality, Pointe-de-Rozel are also definitely of Cambrian age. The Pointe-de-Rozel strata are stratigraphically slightly higher than the Dennemount quarry section. The difference in strike and dip of the two regions provides a possible fold test of remanence age.

Sampling and palaeomagnetic technique

All samples in this study were collected by field drilling; orientation was by sun and/or magnetic compass; magnetic remanence was measured on PAR-SM1 and Schonstedt DSM magnetometers. At each sampling locality at least seven samples were collected from a restricted stratigraphic range. Hence possible secular variation effects were not necessarily annulled. Eight sites were sampled at the Lower Cambrian section at Cartaret, while at the slightly younger Pointe-de-Rozel section three sites were occupied.

Step-wise af demagnetization up to a maximum of 100 mT was applied to at least two specimens from each site. Subjectively, an optimum cleaning range which varied for each site was selected from the information provided by the two pilot specimens. All remaining

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specimens were then treated over the observed optimum range. After this initial af cleaning many specimens were then treated by either thermal demagnetization or chemical leaching. Cancellation of the present Earth's field over the region of the demagnetization furnace was provided by a nest of Schonstedt magnetic shields. In an attempt to further randomize any remanence contamination during thermal treatment, the specimens were rotated about a horizontal axis while cooling down to room temperature. To promote acid leaching the specimens were serrated using a ⅜ in. diameter thin wall drill (Morris & Carmichael 1978). Two modes of leaching were adopted. Both involved the immersion of specimens in 12N HCl acid. One method was at atmospheric pressures, while in the other case the specimens were kept in a partial vacuum of approximately 7 kPa during the leaching period.

To calculate the site mean NRM directions a simple two-tier hierarchical system was adopted. A slight modification of this system was applied to the treated remanence data. Individual remanence direction observations were made in two ways, involving (a) direct measurements, and (b) vector subtractions. In cases where the demagnetization of a specimen yielded a number of estimates (A, B, C) of a removed vector (as calculated by vector subtraction), the specimen direction employed in the hierarchical system is the vector resultant (A + B + C) of the individual vector estimates.

Table 1. Cambrian red beds – Normandy – NRM.

Site	*N* (specs)	*R*	*k*	α_{95}	*D* (*in situ*)	*I*	*D* (dip corrected)	*I*
Dennemount Quarry, Cartaret*								
4	5	4.724	15	21	262	+ 79	201	+ 75
3	6	5.836	31	12	253	+ 66	230	+ 63
8	6	5.960	128	6	261	+ 84	189	+ 79
7	5	4.930	57	10	214	+ 81	179	+ 73
2	6	5.917	61	9	288	+ 66	261	+ 72
1	6	5.932	74	8	214	+ 61	198	+ 46
5	6	5.933	75	8	256	+ 57	234	+ 52
6	6	5.952	105	7	235	+ 58	218	+ 57
Pointe-de-Rozel								
9	6	5.729	18	16	205	+ 6	206	+ 29
10	8	7.770	30	10	225	+ 53	258	+ 81
11	6	5.925	66	8	194	+ 9	192	+ 32

NRM – overall analysis

	N	*R*	*k*	α_{95}	*D*	*I*	Lat.	Long.	$d\psi$	$d\chi$
Dennemount Quarry Cartaret										
dip corrected	8 (sites)	7.743	27	11	216	+ 66	334° E	18° N	15	18
in situ	8 (sites)	7.789	33	10	247	+ 70	319° E	32° N	15	17
Pointe-de-Rozel										
dip corrected	20 specs	16.780	5.9	14	202	+ 49	339° E	4° S	12	19
in situ	20 specs	17.625	8.0	12	207	+ 26	331° E	18° S	7	13

*Sites in ascending stratigraphic order.

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Palaeomagnetic results

All sites have significantly grouped NRM directions. In most cases the site mean directions are well defined only at site 4 was α_{95} greater than 20° (Table 1). In all cases step-wise af demagnetization produced a change in remanence direction towards shallower inclinations. Concomitant changes of magnetic intensity were variable. After 80 mT some pilot specimens (sites 4 and 11) retained over 90 per cent of their original remanence; for the majority between 60 and 70 per cent was retained. Notable exceptions are the pilot specimens from

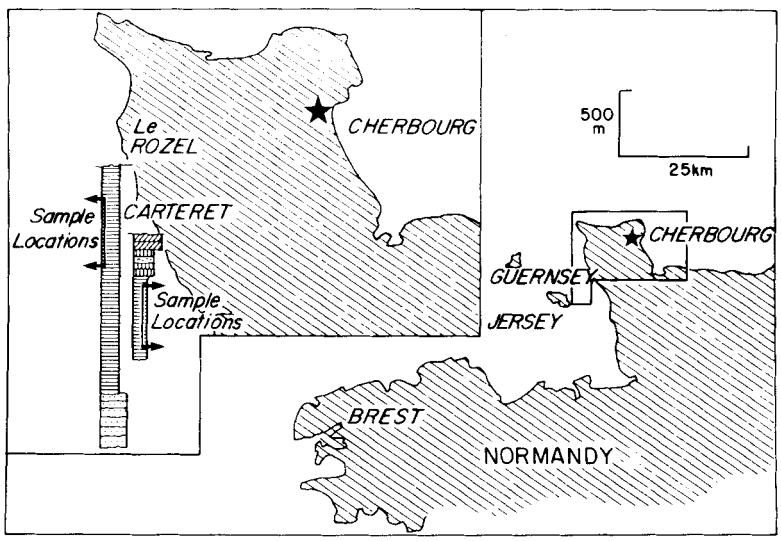


Figure 1. Location of the two sampling areas at Pointe-de-Rozel, and Cartaret. The approximate stratigraphic columns provide an estimate of the relative stratigraphic positions of the two sets of sites. The geological information is based on Doré (1963).

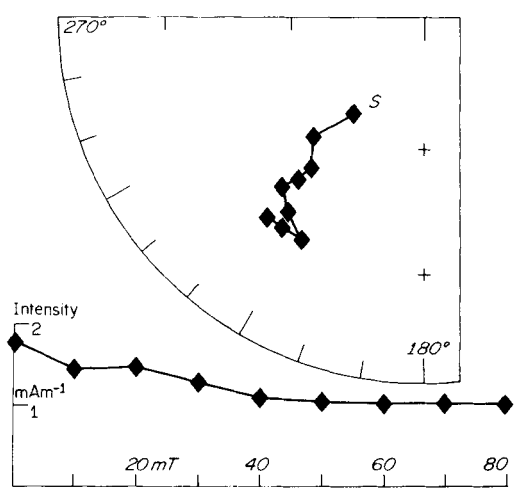


Figure 2. Example of a specimen (2–4) af demagnetization where a distinct trend can be observed from a steep to a more shallowly inclined south-westerly vector. The trends of direction change upon demagnetization also suggest that another NW–SE axis of magnetization may also be present in the data. Because of inaccuracies in the vector subtraction method, and the limited number of determinations no reliable estimate of this NW–SE vector axis could be obtained. Full symbols on the stereonet represent positive magnetic inclination.

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site 10 in which only 10 per cent remained after 100 mT. In a few rare examples two quite distinct remanence components could be identified. For example, Fig. 2 shows the gradual migration of the remanence vector from a steep to a much shallower south-westerly declination. Decomposition of this demagnetization curve by vector subtraction suggests that we are dealing with two remanence components both having south-westerly declinations. It is not possible to make any generalities about the coercivity spectrum of the two remanence components as this varies from specimen to specimen. Suffice to say that in the majority of cases af cleaning did not provide an effective means of vector separation. After this initial cleaning all sites had statistically significantly grouped remanence (Table 2). In some cases cleaning produced a significant increase in site mean precision; most notable is site 3 where k (Fisher's (1953) precision parameter) increased from 31 to 353. Although, the individual site means were well-defined, the site directions did not exhibit any significant clustering. In fact the site directions plotted as a smeared distribution that appeared to crudely fall in stratigraphic succession from the stratigraphically lower sites 4, 3, 7 and 8 to the much higher sites 9 and 11 (Fig. 3). Secular variation does not offer a viable explanation for this smeared distribution. Estimates of secular variation calculated by using the statistical methods described by Gough, Opdyke & McElhinny (1964) or Cox (1969) yield values far

Table 2. Cambrian red beds – Normandy – af cleaned.

Site	Optimum field (Oe)	N	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
Dennemount Quarry, Cartaret*									
4	100	5	4.911	45	12	261	+ 69	225	+ 69
3	300	6	5.986	353	4	257	+ 58	239	+ 58
8	200	6	5.931	72	8	245	+ 77	205	+ 72
7	200	5	4.970	135	7	238	+ 75	208	+ 69
2	400	6	5.855	35	12	239	+ 52	226	+ 50
1	200	6	5.929	71	8	220	+ 46	208	+ 35
5	300	6	5.906	53	9	257	+ 47	241	+ 44
6	500	6	5.924	66	8	222	+ 34	215	+ 32
Pointe-de-Rozel									
9	300	6	5.943	87	7	194	0	194	+ 22
10	500	8	7.772	25	11	210	+ 6	208	+ 37
11	300	7	6.908	66	8	194	−1	194	+ 22

af cleaned – overall analysis

	N	R	k	α_{95}	D	I	Lat.	Long.	$d\psi$	$d\chi$
Dennemount Quarry, Cartaret										
dip corrected	8 (sites)	7.660	21	13	222	+ 54	325° E	6° N	12	18
in situ	8 (sites)	7.672	21	12	240	+ 58	314° E	17° N	13	18
Pointe-de-Rozel										
dip corrected	21 spec	20.295	28	6	199	+ 28	339° E	18° S	4	7
in situ	21 spec	20.341	30	6	199	+ 2	335° E	32° S	3	6

* Sites in ascending stratigraphic order.

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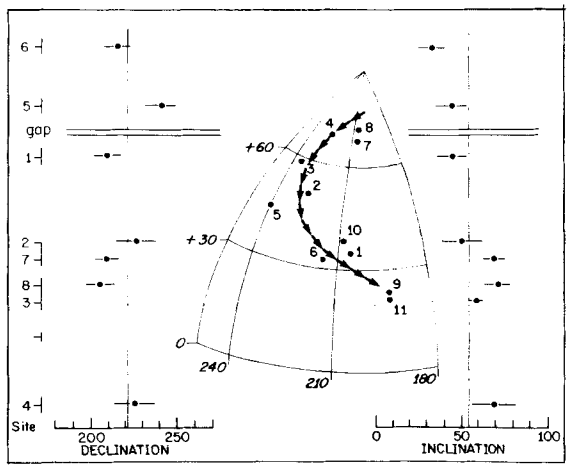


Figure 3. Site mean directions of magnetization after af treatment plotted with respect to their stratigraphic position. The error bars represent the α_{95} confidence interval. The broken lines are the overall mean declination and inclination. Inset is an equatorial stereographic projection of the same site mean data. The lower stratigraphic sites have steep inclination, the higher sites have shallow inclination, with the remainder showing a gradual progression between the two extremes.

in excess of that expected for the specified palaeolatitude (Irving 1964). Explanations for the directional smearing are to be sought in either partial remagnetization, or rapid polar shift during sediment deposition. For the former explanation to be valid we accept that some of the initial remanence has been masked by a secondary component that was probably chemically imprinted on the sediment at some instantaneous later date. On this model, the total remanence acquisition of a red bed is considered to be a long term process (Roy & Park 1974). That is not to say, however, that it is also a time averaging process. Critical to this point is the rate of secondary (colloidal) haematite deposition, and especially the rate at which individual grains grow through the critical remanence blocking threshold. Haematite may be depositing over a long period, but the grains only record stable remanence after passing the critical blocking size and this may take place over a very short time period. Studies on other red beds (Morris 1978; Roy & Robertson 1978; Morris & Carmichael 1978) where particular directions can be associated with specific haematite phases suggests this blocking occurs at several instances of time, possibly related to specific dewatering (crystallization) events. Continuous formation and remanence blocking of smaller quantities of haematite probably begins straight after deposition associated with the *in situ* chemical degradation of iron rich minerals. This phase could possibly not produce a meaningful palaeomagnetic signal (Larson & Walker 1975) depending upon the time required to chemically degrade all the iron rich minerals that are present. Only the intervals related to significant haematite depositing events will be amenable to later detection.

In contrast, for the second alternative (true polar wander) to be valid, we require that in each case the observed remanence be initial (primary?). This requires that the total remanence of a specimen be acquired quickly and that subsequently no further haematite deposition contaminates this initial remanence. As noted already a few examples of the af data can be resolved into two distinct remanence components. To further investigate this suggestion of multi-phase remanence, thermal treatment and acid leaching were applied to the same specimens.

Thermal cleaning was applied in steps of 50°C up to a maximum of 650°C. In the majority of cases this treatment also did not reveal any systematic underlying secondary

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components, little change in direction being observed until after 550°C, when less than 50 per cent of the post af cleaning intensity remained. The few specimens that did show a high unblocking temperature component exhibited a slow change toward shallower inclinations with a south-westerly declination. Further thermal demagnetization produced anomalous direction and intensity changes. Rapid increases of intensity coincided with a marked visible reddening of the specimen. Slicing showed this effect was restricted to the surface of the specimen. Polished section examinations revealed that this effect involved the dehydration of mica grains and their subsequent oxidation to produce a new generation of oxide minerals, most of which appeared to be haematite. Hence, when these specimens were exposed to the local Earth's field, they were capable of acquiring a new remanence component. Commonly this seems related to placing the specimens in the specimen holders. Further evidence of this laboratory induced effect is provided by the magnetic directions. Whilst, within site grouping of the *in situ* results significantly decreased with increasing oxidation, the precision of the relative to core directions became very tighly grouped. That is, we observe that the remagnetized direction preferentially lies along the length of the core. Attempts were made to remove this contamination direction using acid leaching. In most cases these were unsuccessful. Fig. 4 shows two examples of acid leaching on thermally contaminated specimens. One specimen, Fig. 4(a), is from the stratigraphically low site 3. Fig. 4(b) is the highest stratigraphic level, site 11. Both specimens were treated by all three cleaning methods, first af, then thermal, and finally by acid leaching. In both instances the af treatment produced a shift towards the shallow south-westerly direction. Initial thermal treatment did not produce any significant direction changes, lending support to the af determined points as meaningful remanence directions. Further thermal treatment produced a shift towards steep inclinations. Acid treatment was then commenced, and this was associated with a marked rise of remanent intensity for both samples.

For the specimen in Fig. 4(b), this intensity increase was associated with a shift back down to shallow inclinations. Note, however, that the declination appears to be somewhat

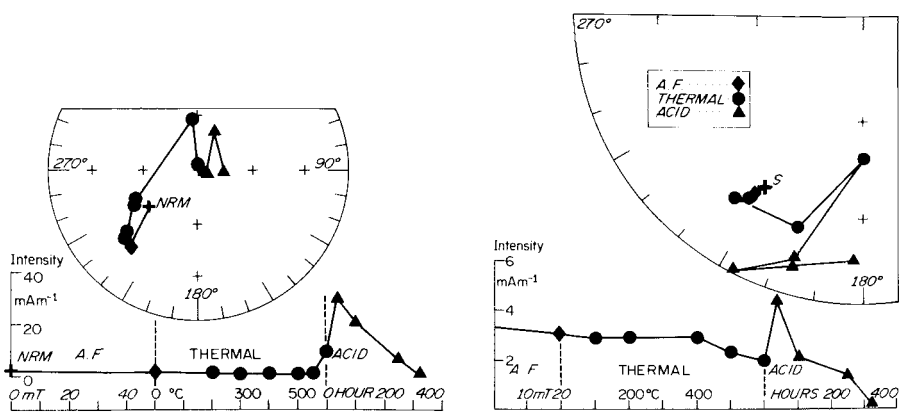


Figure 4. Stereonets and intensity decay curves to show the effects of acid leaching upon two specimens, from high (b) and low (a) stratigraphic levels that have been contaminated by thermal cleaning. Both specimens (a) (specimen 3–1) and (b) (specimen 11–2) show a small shift to shallower inclinations upon af demagnetization, and remain there for the initial steps of thermal demagnetization. Upon reaching higher temperatures, specimen (a) shows a shift towards steeper inclination, together with an increase of remanent intensity. This intensity continues to increase with initial acid leaching, before it begins to decay. (b) Shows a similar shift to steep inclination, but the intensity increase only begins with acid cleaning. The remanence of (a) never moves back down to shallow inclinations, whereas that of (b) does. From this it would appear that it is more possible to remove the contaminant remanence carrying phase from the stratigraphically higher sites than from the lower sites.

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different. It is uncertain whether this represents another distinct direction or not. From this collection then it appears that once a specimen has been contaminated by heating above 600°C then the remanence of the contaminated grains continues to dominate all further remanence measurements. It appears that the magnitude of the remanence due to the contaminated phase is far in excess of any other remanence components the specimens may possess. Hence, even if acid leaching did preferentially remove other remanence components their effect would probably be unobserved. Further compounding this problem, the contamination direction is the most resistant to acid leaching, possibly because in some instances the haematite phases are located within silicate grains. In summary then, once remanence contamination produced by thermally induced chemical alteration has been initiated then further retrieval of meaningful data becomes next to impossible.

By far the most successful method of separating the two remanence components was simple acid leaching with only prior af cleaning. Yet even this method failed to work on specimens from some sites. In the cases where it worked, the revelation of the underlying remanence was quite rapid, usually requiring less than 100 hr (Fig. 5). Further leaching of the specimens merely produced a gradual decay of intensity until fragmentation of the specimens prevented any further measurements. For those specimens in which acid leaching did not uncover a second magnetization, the limitation on the extent of acid leaching was imposed by the measuring limit of the magnetometers due to the decrease in intensity of magnetization with prolonged leaching. Hence it is possible that with a more sensitive magnetometer acid leaching could eventually have uncovered the second magnetization even in these specimens but in the present study the direction could not be determined with any accuracy.

Results

Table 3 lists the two remanence directions as they have been determined at each of the stratigraphic sampling levels. All sites, except site 9, appear to carry at least some of the steeply inclined FA direction. The relative contribution of this component to the total magnetic vector of an average specimen varies up the stratigraphic column. No trace of any

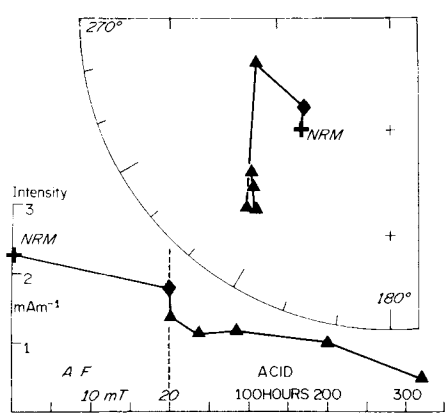


Fig. 5. Representative specimen to show the effects of acid leaching (represented by the triangles) after af cleaning (shown by the diamond). The first chemical step represents the effect of drilling a central hole in the specimen. The stereonet shows that for this particular specimen (6–3) the shallow direction is observed after only a mere 50 hr of leaching. Further treatment produces only a gradual decay of intensity, until after 328 hr no further treatment was possible because the specimens had become too friable.

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Table 3.

Site	Magnetization A								Magnetization B							
	N	*D* (*in situ*)	*I* (*in situ*)	*D* (tilt corrected)	*I* (tilt corrected)	*R*	*k*	α_{95}	*N*	*D* (*in situ*)	*I* (*in situ*)	*D* (tilt corrected)	*I* (tilt corrected)	*R*	*k*	α_{95}
Cataret																
4	5	261	69	225	69	4.911	45	12								
3	6	261	62	240	63	5.971	174	5								
8	6	258	78	210	74	5.971	170	5								
7	5	240	73	208	69	4.971	138	7								
2	5	272	68	253	66	4.932	59	10	5	204	32	201	23	4.89	38	13
1	no reliable determinations								5	213	30	209	24	4.94	67	9
5	4	281	78	233	71	3.96	67	11	6	237	28	230	18	5.87	40	11
6	6	298	75	256	82	5.94	86	7	6	209	18	206	18	5.82	29	12
Pointe de-Rozel																
9									6	203	07	202	27	5.94	85	7
10	8	226	83	006	72	7.79	33	10	7	205	−5	203	23	6.86	42	9
11	6	174	64	110	73	5.79	24	14	7	200	−2	199	22	6.81	31	11
Overall means																
in situ	9	253	75			8.78	40	8	7	210	16			6.66	18	15
tilt corrected	9			230	78	8.68	25	11	7			207	22	6.90	61	8

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other direction is observed at the lower sampling sites; the FA direction is totally dominant. At the highest stratigraphic levels, especially the sites from Pointe-de-Rozel, the FA direction is seen as a weak overprint that can easily be erased by either thermal cleaning or acid leaching. Determination of the FA direction at these stratigraphically high sites is based on vector subtractions of clearly defined FB directions from NRM directions.

Critical to this kind of analysis is the assumption that for the interval over which the vector subtraction is effected, the removed remanence should be totally dominated by the component of which one is attempting to obtain a directional estimate. Should some demagnetization of another component also take place during this interval, then clearly the directional estimates obtained will exhibit some systematic bias away from the true direction. Where direct measurements of the vector subtraction resolved component are absent, it is not possible to determine how much bias has been introduced. For the present data set there does not seem to be any systematic difference between the higher stratigraphic sites where the FA directions are based on vector subtractions, and the low stratigraphic site FA directions that are based on direct observations. This suggests little contamination in these vector subtractions, and hence little bias to the overall mean direction.

Applying the same logic to the FB directions, we see that it is sites 2, 1, 5 and 6 where a vector contamination problem may arise. Applying the Watson–Irving test (1957) to this data set suggests that the site 5 mean direction is significantly distinct from the other site means, and therefore may contain a significant proportion of unresolved FA component. Including the site 5 mean in the total produces little difference in the overall mean direction, from $D = 207^\circ$, $I = +22^\circ$ and $\alpha_{95} = 8^\circ$ with site 5, to $D = 203^\circ$, $I = +22^\circ$, $\alpha_{95} = 4^\circ$ without site 5. Hence for the remainder of this paper we will consider the overall mean including site 5.

In any study where two (or more) remanence directions are identified it is important that the correct sequence of remanence acquisition be defined. For this particular study there are three lines of evidence to suggest that the sequence was FB–FA. By far the most powerful evidence is provided by a fold test. This test is facilitated by the difference in amount and direction of dip of the strata between Pointe-de-Rozel and Dennemount Quarry, Cartaret. Comparing the Fisher precision of the tilt corrected and *in situ* mean FB directions produces an increase which is significant at the 99 per cent level (McElhinny 1964); it may be inferred that the FB direction predates folding of the strata. Applying the same tilt corrections to the FA data produces a small decrease of Fisher precision, merely suggesting (but not proving) that the FA remanence postdates folding. The second line of evidence comes from acid leaching. Throughout all the leaching experiments FA is preferentially erased and FB is revealed. If one then assumes that the original detrital grains offer the greatest resistance to leaching then it follows that the last direction revealed by leaching must have been the first acquired (Roy & Park 1974).

Finally, this sequence is also supported by polished thin section examinations. Plate 1 shows photomicrographs of typical specimens from the lower sites dominated by A remanence (Plate 1a), and the upper sites (especially 9 and 11) dominated by B remanence (Plate 1b). Both photomicrographs were taken at the same scale with a mixture of reflected and transmitted light under partially crossed nicols. Plate 1(a) shows a distinct planar fabric delineated by large amounts of secondary amorphous haematite. Little primary detrital magnetic material is visible, but some minute grains exhibit later haematite overgrowths. Plate 1(b) in contrast shows an extremely small amount of secondary haematite; detrital haematite grains are commonly seen and overall the rock exhibits no significant tectonic texture, except that possibly induced by compaction. Again these features also favour the magnetization sequence FB–FA.

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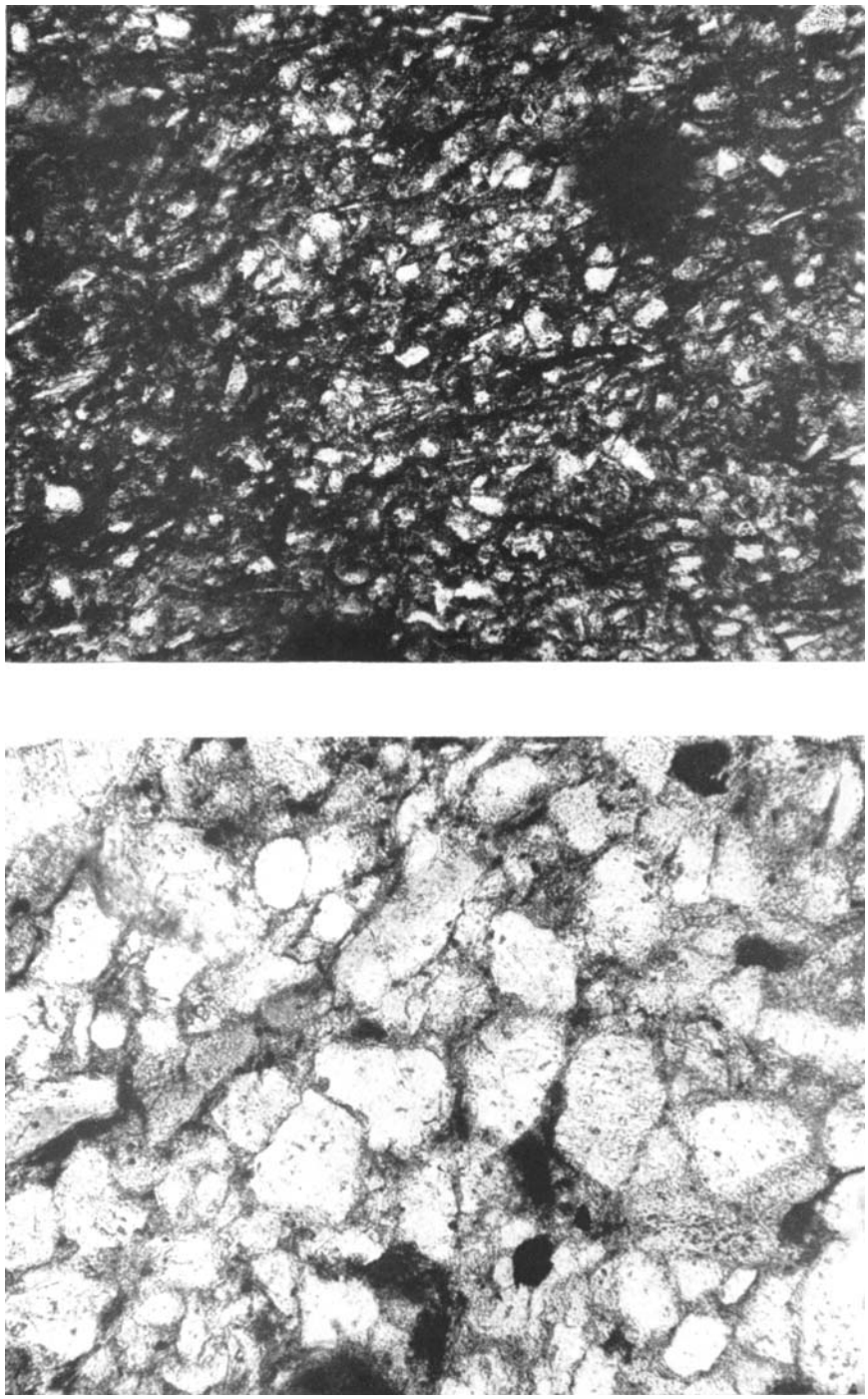


Plate 1. Photomicrographs of typical specimens from (a) the lower sites dominated by the A remanence, and (b) the upper sites dominated by the B remanence. Both photomicrographs were taken at the same scale with a mixture of reflected and transmitted light under partially crossed nicols.

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Discussion

Detailed magnetic cleaning and analysis of this Lower Cambrian sequence of red beds has shown that an apparent stratigraphic smear of remanence directions actually records the complex interplay of two remanence components. One of these directions (FB, $D = 207°$, $I = 22°$, Pole 13° S, 301° E, $\alpha_{95} = 8°$), which definitely predates folding of the strata, probably dates from the initial depositional event. The age of the second magnetization (FA, $D = 253°$, $I = 75°$, Pole 22° N, 354° E, $\alpha_{95} = 11°$) is less secure, but a lower Ordovician erosional event provides a possible mechanism.

Fig. 6 compares these newly acquired data with other published Cambrian and Ordovician data from Central Europe and the British Isles. Poles from Central Europe are represented by dots when based on single site observations, and squares when based on multiple site observations. These data represent studies by Bucha and his co-workers from the Ukrainian Massif (Bucha 1961, 1965; Andreeva, Bukha & Petrova 1965). At the time of writing no other Lower Palaeozoic poles have been published from Central Europe. Hence it is uncertain whether these data can be considered representative of the whole of Central Europe, or merely the Ukrainian Massif. Until very recently, the apparent separation of these results has been considered indicative of some unexplained aberrancy. Especially, the wide

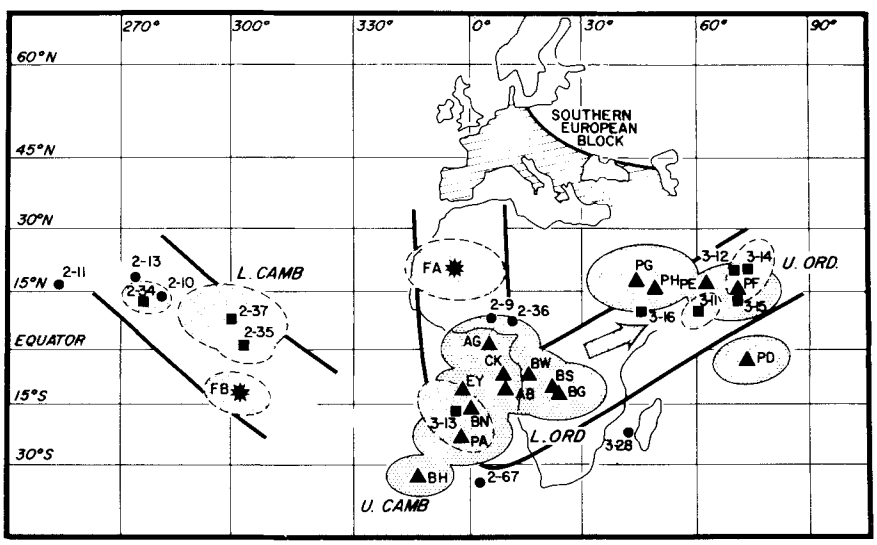


Figure 6. Comparison of poles FB and FA from this study to other Cambrian and Ordovician poles from central Europe (dots and squares), and Britain (triangles). The central European poles are identified by their Ottawa catalogue number (Irving *et al.* 1976). The squares represent collections based on more than one site. The dots represent data from one site; mostly these are from the Barrandian Porphyries of Czechoslovakia. The British poles are: BG – Ballantrae Gabbros, BS – Ballantrae serpentinites, BH – Bail Hill lavas, AG – Ashgill intrusives, AB – Aberdeenshire Gabbros, BN – Builth lavas, EY – Eyott lavas, CK – Carrock Fell intrusives, BW – Borrowdale lavas, BX – Breidden Hill extrusives, BY – Breidden Hill intrusives, the poles PA, PD, PE, PF, PG, PH are from Piper's (1978) Table 5. Pole FB, the initial remanence direction from these Lower Cambrian red beds plots slightly to the south of similar age material from central Europe. Because of the limited data base no tectonic importance can be attached to this minor discrepancy. Little is known about the age and significance of the secondary magnetization pole FA except merely that it probably postdates folding which is locally of Middle to Upper Cambrian age. Possibly it could be associated in time with the local post-Arenig transgression (Doré 1963). The pole path between the Lower and Upper Cambrian poles is not presently defined because of a lack of data.

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variation of directions observed in the Barrandian Porphyries has been somewhat puzzling.

It is clear from more recent experimental work (for example Symons 1978; Palmer *et al.* 1979), that patchy metasomatism, or metamorphism, or even unidentified subsequent intrusion may produce distinct directional groupings in different segments of an intrusion. In recognizing this feature, we have chosen to plot individual site poles from these studies. Having done this some appear to correspond with Lower Cambrian poles, while others are closer Ordovician ones. All the British poles which are based on a number of sites, are of Ordovician age. No British Cambrian poles are plotted since their proximity to the Carboniferous poles renders them questionable.

No meaningful comparisons, then, are possible between Central Europe and Britain for any portion of the Cambrian. Moreover, with the limited data, it is also not possible to determine whether the small discrepancy between pole FB, and the other Lower Cambrian poles is of any tectonic significance. At present, therefore, we cannot detect whether there was any motion between the various elements of Central Europe (Lorenz 1976), or even between Central Europe and Britain in the Cambrian. The available poles form a crude swath indicated by the parallel lines, this may possibly represent a segment of a major Cambrian Loop. Data for the same period from the Gondwanic continents, which add support to this interpretation (Morris & Carmichael 1978), define a crudely subparallel pole path trending in age sequence from south to north of the equator. The connection between the Lower and Upper Cambrian data for the Gondwanic continents is achieved by a high latitude northerly loop. No evidence for such a loop is yet known from Central Europe. To depict this uncertainty the path between the Cambrian and younger data is not defined on Fig. 6, and its definition must await the publication of other Cambrian data from Central Europe.

Pole FA plots slightly to the north of a group of middle Ordovician poles that includes data points both from Central Europe and Britain. This minor discrepancy could have a number of explanations. It must be remembered that we have merely surmised an Ordovician age for this pole. Accepting a younger, possibly Devonian, age may provide a better match (Creer 1968). The location of the pole prohibits any age younger than Carboniferous, since by this time the pole had begun to migrate towards higher latitudes as defined on the present geographic grid. Alternatively if we accept the Ordovician age, then it is possible to construct a polar sequence from pole FA through the group of middle Ordovician poles to the group of upper Ordovician poles (Fig. 6). On this interpretation pole FA would probably be of basal Ordovician age. This accords quite nicely with the recognized Arenig erosional event, which terminated the Cambrian depositional sequence in northwestern France. On this evidence, therefore, we prefer a lower most Ordovician age for pole FA.

Palaeomagnetic evidence, initially presented by Briden, Morris & Piper (1973), and later substantiated by Piper (1978) has shown that during the interval Lower to Middle Ordovician central Britain/southern Ireland was at its greatest distance from northern Europe. The data depicted in Fig. 6 suggest that during this same time period central Britain and southern Europe behaved as a unified entity. This accords well with the faunal evidence for these two regions belonging to the same faunal province during the Lower Ordovician (Whittington & Hughes 1972; Burrett 1973). Data are not yet available to show if this coincidence of pole location applied prior to and after this Lower Ordovician interval. Hence as yet it is uncertain if Britain really did move to and fro across a mid-European ocean. Finally, the coherence of this common pole path segment indicates that there was no relative rotation at least between central Britain and the Bohemian massif, during the development of the Hercynian or Alpine orogenies.

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Acknowledgments

Professor R. M. Shackleton noting the importance of palaeomagnetic studies in central Europe provided the initial impetus to collect these samples. His geological assistance in the field was also appreciated. Initial measurements were made at the University of Leeds, England while an NERC graduate student and final measurements were made at the University of Western Ontario, London, Ontario, Canada, while in receipt of a NRC post-doctoral fellowship.

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